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# Memory Overload

EVOLUTION DID NOT ENDOW HUMANS with the ability to play pick-up basketball. True, it produced legs for running, hands for dribbling, and shoulders for fouling, but all that this enables us to do is shoot hoops by ourselves. To get into a game with the strangers we find in the schoolyard on any given afternoon, we not only have to work in concert with four teammates we may never have met before—we also need to know that the five players on the opposing team are playing by the same rules. Other animals that engage strangers in ritualized aggression do so largely by instinct—puppies throughout the world have the rules for rough-and-tumble play hard-wired into their genes. But American teenagers have no genes for pick-up basketball. They can nevertheless play the game with complete strangers because they have all learned an identical set of ideas about basketball. These ideas are entirely imaginary, but if everyone shares them, we can all play the game.

The same applies, on a larger scale, to kingdoms, churches, and trade networks, with one important difference. The rules of basketball are relatively simple and concise, much like those necessary for cooperation in a forager band or small village. Each player can easily store them in his brain and still have room for songs, images, and shopping lists. But large systems of cooperation that involve not ten but thousands or even millions of humans require the handling and storage of huge amounts of